CS 252:

Advanced Programming Language Principles

Typed Arith

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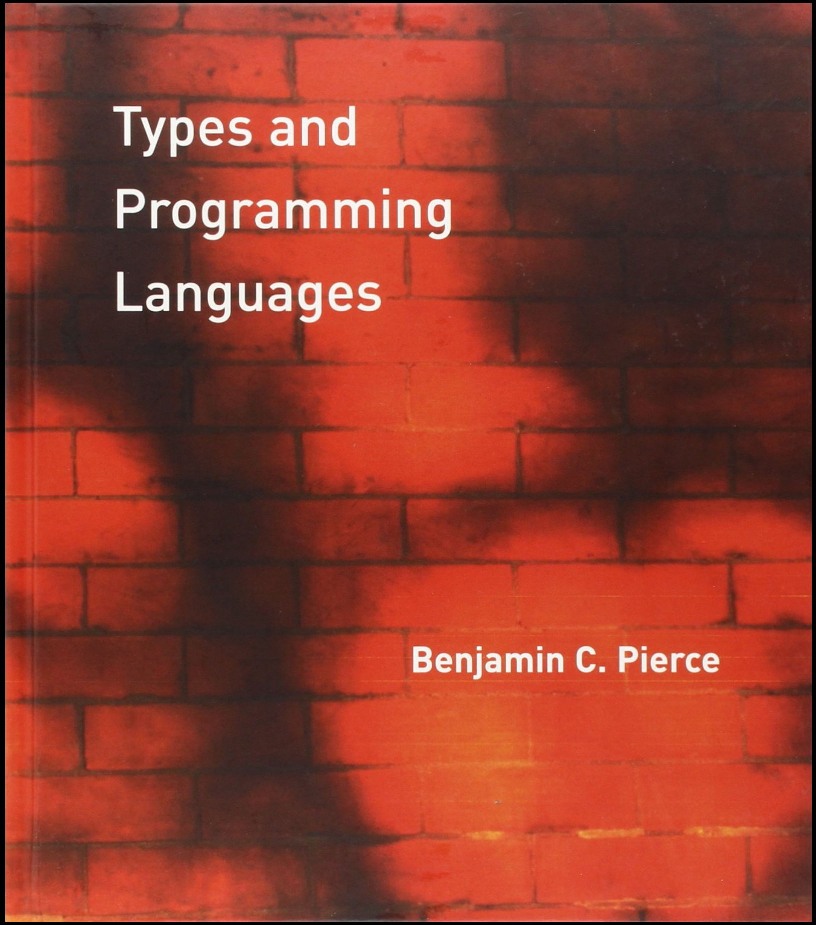
Types & Programming Languages (TAPL)

- Standard reference
- Copies available at the library
- Chapter 8



Arith Language

```
e ::= true
    | false
    | n
    | succ e
    | pred e
    | iszero e
    | if e then e else e
```

```
v ::= true
    | false
    | n
```

Small-step evaluation rules for Arith
(in-class)

Types for Arith

Our typing rules will be of the form

$$\texttt{e:T}$$

This says that an expression `e` will either

1. evaluate to a value of type `T`, or
2. go into an infinite loop.

Types for Arith

```
T ::= Bool
    | Int
```

Typing rules for Arith
(in-class)

Is our type system "good"?

- Does it catch "bad" programs?
- Are there "good" programs that it prevents us from running?
- And what do we mean by "good" and "bad"?

Type Safety

If an expression *typechecks,* then it won't "get stuck". Either:

- the expression is a value
- an evaluation rule reduces the expression to a different expression

Safety = Progress + Preservation

- **Progress**: A well-typed expression won't get stuck.
- **Preservation**: Evaluating a well-typed expression will result in a well-typed expression.

Type Safety, Formally

- Progress theorem:
  Suppose `e:T`. Then either
  1. `e` is a value
  2. There exists an `e'` such that `e -> e'`
- Preservation theorem:
  Suppose `e:T` and `e -> e'`.
  Then `e':T`.

Lab: Extended Arith Language

```
e ::= …
      | s
      | concat e e
      | isemptystr e
      | strlen e
```

```
v ::= …
      | s
```

```
T ::= …
      | String
```